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United States Patent	12404956
Kind Code	B2
Date of Patent	September 02, 2025
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Pipe connection arrangement for a fluid management system

Abstract

A pipe connection arrangement for a fluid management system includes a pipe nut for releasably connecting an end portion of a pipe to a fluid port of the fluid management system. The pipe nut has a first portion having a toothed outer profile and a second portion configured to engage with a connection point portion of said fluid management system. An elongated drive member having a first end portion with a toothed outer profile is configured to cooperate with the toothed outer profile of the pipe nut, the first end portion further being configured to concentrically engage with a supplementary guide member adapted to be fixedly arranged adjacent the connection point portion of said fluid management system. When the elongated drive member is in engagement with the supplementary guide member, and the second portion of the pipe nut is in engagement with the connection point portion, the toothed outer profile and the toothed outer profile of the pipe nut are permitted to mesh together such that a rotation of said elongated drive member causes a corresponding rotation of the pipe nut for clamping the end portion to the fluid port or unclamping the end portion from the fluid port.

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Appl. No.:	18/342921
Filed:	June 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240011584 A1	Jan. 11, 2024

Foreign Application Priority Data

EP

22183885

Jul. 08, 2022

Publication Classification

Int. Cl.: F16L19/00 (20060101); F02M61/14 (20060101); F16L19/065 (20060101)

U.S. Cl.:

CPC F16L19/0653 (20130101); F02M61/14 (20130101); F02M2200/803 (20130101)

Field of Classification Search

CPC: F16L (19/0653); F02M (61/14); F02M (61/168); F02M (2200/803); F02M (55/02); F02M (37/0017); B25B (15/005); B25B (27/0035); B60T (17/043)

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a pipe connection arrangement for a fluid management system. In particular, the disclosure relates to a fuel pipe connection arrangement for a fuel management system of a vehicle, wherein the fuel management system comprises a fuel injector. The disclosure also relates to a fluid management system for a vehicle, in particular a fluid management system comprising a fuel injector. Moreover, the disclosure relates to a vehicle comprising such a pipe connection arrangement and/or fluid management system. The disclosure is applicable on vehicles, in particular heavy-duty vehicles, such as trucks, buses and construction equipment. Although the disclosure will be described in relation to a truck, the disclosure is not restricted to this particular type of vehicle, but may also be used in other types of vehicles such as cars, working machines within the fields of industrial construction machines or construction equipment, such as articulated haulers, dump trucks, wheel loaders and the like. The disclosure may also be used in other types of industrial machines, engines, marine vessel and applications.

BACKGROUND

(2) In fluid management systems for vehicles, such as fuel systems for engines, there is a need for installing new components and performing service maintenance on a regular basis in order to ensure reliable and long-lasting operations. Such operations may generally include or involve installations of pipes, such as connecting or disconnecting one or more fuel pipes to one or more fuel injectors of a fuel management system.

(3) One challenge during installation of pipes is the lack of space for any needed tightening tools when pipes are installed in spaces where there is limited sideways access to the pipe connection point. One example of such environment in a vehicle is the injector installation environment under a valve cover, or a rocker cover, and between the engine valve springs of an engine. Typically, the fuel pipes may need to be connected to the injector with their connection points positioned in between the springs with limited space around, where a pipe additionally forms a bend relatively close to its connection point thus preventing access for the tightening tool. Another example of such an installation may be a pneumatic variable valve actuation system, where complex assemblies are installed in a restricted space under the rocker cover.

(4) There is thus a desire to provide an improved pipe connection arrangement for a fluid management system. In particular, there a desire to provide an improved pipe connection arrangement for a fluid management system of a vehicle.

SUMMARY

(5) It is an object of the present disclosure to facilitate installation of a pipe to a fluid management system, such as fluid management system of a vehicle.

(6) According to a first aspect of the present disclosure, there is provided a pipe connection arrangement for a fluid management system. The pipe connection arrangement comprises a pipe nut for releasably connecting an end portion of a pipe to a fluid port of the fluid management system. The pipe nut comprises a first portion having a toothed outer profile and a second portion configured to engage with a connection point portion of the fluid management system. The pipe connection arrangement further comprises an elongated drive member having a first end portion with a toothed outer profile. The toothed outer profile is configured to cooperate with the toothed outer profile of the pipe nut. The first end portion is further configured to concentrically engage with a supplementary guide member adapted to be fixedly arranged adjacent the connection point portion of the fluid management system. Hereby, when the elongated drive member is in

engagement with the supplementary guide member, and the second portion of the pipe nut is in engagement with the connection point portion, the toothed outer profile and the toothed outer profile of the pipe nut are permitted to mesh together such that a rotation of the elongated drive member causes a corresponding rotation of the pipe nut for clamping the end portion to the fluid port or unclamping the end portion from the fluid port.

(7) By providing a pipe nut for releasably connecting the end portion of a pipe to a fluid port of the fluid management system, the arrangement is configured to clamp and retain, in a fluid-sealing manner, the pipe to the fluid port of the fluid management system.

(8) In this manner, the proposed pipe connection arrangement contributes to an improved functional interface between the pipe and the fluid management system. The present disclosure is based on the insight that generous space allowances are needed for the design of components with pipe connections making up a particular assembly unit, to ensure pipes can be installed and tightened/removed with acceptable difficulty. This usually involves providing the necessary minimum space for tightening tool access, causing the overall design to become relatively complex, bigger in size etc. Occasionally, the components of the assembly may be designed more compact at the cost of having to design and use special tools that can fit tighter spaces for pipe tightening in each particular installation. In other cases, special clamps may be utilized instead of the pipe nuts. This type of arrangement may allow for more compact designs, but generally at the expense of increasing the separate parts' count, introducing loose parts (screw, clamp) that could fall away and/or be forgotten inside the engine after assembly. Still, the use of a clamp may generally require an appropriate space and may also complicate the design of matching components.

(9) Moreover, the proposed pipe connection arrangement provides for pipe-co-axial access of the pipe nut tightening tool, thus allowing for pipe installations of such configurations that may otherwise be difficult in e.g. a common engine environment of a heavy-duty vehicle because of the space constraints. The proposed pipe connection arrangement further allows for using standard tools, whilst reducing the need for using additional separate parts such as clamps and screws.

(10) It should be noted that the fluid may refer to any type of gaseous and liquid fluid medium. By way of example, the fluid is a fuel. The fuel may be provided in gaseous phase or liquid phase. The fluid may also be a liquid such as oil for the engine, e.g. lubrication oil. However, the liquid may also be transmission oil, oil for lubricating a shaft, or a fluid in a cooling system, a working fluid in a braking system, a working fluid in a hydraulic power transmission and/or a working fluid in a control system etc.

(11) While the pipe connection arrangement can be used for various types of fluid management systems so as to connect a pipe to a component of the fluid management system, the example embodiments of the disclosure are particularly useful for a fluid such as a fuel, e.g. hydrogen fuel. Hence, according to one example embodiment, the pipe connection arrangement is a pipe connection arrangement for connecting a hydrogen fuel pipe to a fluid management system in the form of an injector assembly of an engine, e.g. an internal combustion engine of a vehicle.

(12) It should also be readily appreciated that the elongated drive member may only be a part of the pipe connection arrangement during installation and removal of the pipe connection. In this context, the elongated drive member may be considered as a complementary tool for connecting the pipe to the fluid management system.

(13) Typically, when the pipe nut is in threaded engagement with a corresponding threaded part of the connection point portion, an axis of rotation of the pipe nut may essentially be parallel to the centre axis of the supplementary guide member.

(14) The elongated drive member may generally be arranged and configured to engage with the toothed outer profile of the pipe nut.

(15) The first end portion may comprise a recess arranged concentrically about a centre axis of the elongated drive member and further adapted to at least partly encompass an outer end part of the

supplementary guide member. In this manner, the positioning of the elongated drive member relative to the pipe nut may be further improved.

(16) The elongated drive member may comprise a cylindrical recess arranged concentric with the first end portion and extending from the end face of the first end portion. The cylindrical recess and the supplementary guide member may form a locational clearance fit upon concentric engagement of the supplementary guide member and the elongated drive member.

(17) In other examples, the supplementary guide member may be a segment formed by a cylindrical part having an internal diameter. The cylindrical part may be comprised in the fluid management system. The internal diameter of the cylindrical part may be configured such that upon the concentric engagement of the elongated drive member with the supplementary guide member, the internal diameter of the segment forms a locational clearance fit with a maximum diameter of the toothed outer profile.

(18) Typically, when the first end portion of the elongated drive member is engaged with the supplementary guide member, the elongated drive member may be arranged adjacent the pipe nut and corresponding centre axis of the elongated drive member and the pipe nut, respectively, may be arranged parallel to each other. In this manner, the positioning of the elongated drive member relative to the pipe nut may be further improved. As such, the elongated drive member may generally be arranged adjacent the pipe nut as seen in a plane extending perpendicular to corresponding centre axis of the elongated drive member and the pipe nut.

(19) The pipe connection arrangement may typically comprise the supplementary guide member. The supplementary guide member may be positioned at a distance from the connection point portion such that an axis of rotation of the pipe nut, when in engagement with the connection point portion, may be essentially parallel to a centre axis of the supplementary guide member. In this manner, the positioning of the elongated drive member relative to the pipe nut may be further improved. The distance may generally refer to a distance in the transverse direction and/or longitudinal direction.

(20) The supplementary guide member may be arranged at a position adjacent the pipe connection point portion. As such, the supplementary guide member is associated with the fluid port.

(21) By way of example, the supplementary guide member is a guide pin for arrangement in the fluid management system. The supplementary guide member may at least partly be of a cylindrical shape.

(22) The pipe connection arrangement may further comprise an additional supplementary guide member. The additional supplementary guide member may be positioned at an essentially same transverse distance from the connection point portion as the supplementary guide member, but at different angular location around a circumference of the connection point portion relative to the angular location of the supplementary guide member. Hereby, it becomes possible to further improve the installation of the pipe to the fluid management system. Also, the above configuration with the additional supplementary guide member provides for improving the timing of the meshing cooperation between the toothed profiles.

(23) The additional supplementary guide member may be essentially identical in design to the supplementary guide member.

(24) The supplementary guide member may be a first supplementary guide member and the additional supplementary guide member may be a second supplementary guide member. The second supplementary guide member may be arranged asymmetrical from the position of the first supplementary guide member relative to the axis of rotation of the pipe nut.

(25) The additional supplementary guide member may be positioned at an angular location that is not a multiple of 360 degrees divided by a number of teeth of the toothed outer profile. In this manner, it may be possible to use a less complex shaped pattern of the toothed profile that generally require more than one relative position of the driving and driven members for efficient enough force transfer.

(26) The additional supplementary guide member may be positioned at an angle being greater than 120 degrees relative to the supplementary guide member. Such arrangement may at least partly overcome tilting of the pipe nut in its thread that increases friction in the threads and reduces useful torque.

(27) The pipe connection arrangement may comprise an additional elongated drive member having a corresponding first end portion comprising a corresponding outer profile configured to mesh with the toothed outer profile upon engagement with and subsequent rotation of the additional elongated drive member. The corresponding first end portion may be configured to at least partly encompass a corresponding outer end part of the additional supplementary guide member.

(28) The second portion may be an outer threaded profile of the pipe nut. Alternatively, the second portion may be an inner threaded profile of the pipe nut. If the second portion is provided in the form of an outer threaded profile, the second portion may be a part of the outer surface of the pipe nut and arranged space-apart from the toothed outer profile of the first portion.

(29) The toothed outer profile of the elongated drive member may define a six-point star-shaped pattern, such as a TORX-profile. The toothed profile of the elongated drive member may define a standard TORX-profile and the diameter of the cylindrical recess may essentially be equal to a diameter of a central recess of a "Security TORX" screwdriver bit.

(30) The pipe connection arrangement may generally also comprise the pipe. By way of example, the pipe nut may be arranged about a part of the pipe.

(31) According to a second aspect of the present disclosure, there is provided a fluid management system. The fluid management system comprises a pipe connection arrangement according to the first aspect of the disclosure.

(32) Effects and features of the second aspect are largely analogous to those described above in relation to the first aspect.

(33) The supplementary guide member may be a separate part attached to the fluid management system.

(34) The supplementary guide member may be comprised with the fluid management system. The supplementary guide member may be provided in the form of a cylindrical projecting segment. Alternatively, the supplementary guide member may be provided in the form of a cylindrical recessed segment with an inner diameter.

(35) The supplementary guide member may be a cylindrical projecting segment being machined in the fluid management system. The supplementary guide member may be a cylindrical recessed segment being machined in the fluid management system.

(36) The fluid management system may comprise a fuel injector assembly for an internal combustion engine. The fuel injector assembly may be a hydrogen fuel injector assembly configured to inject hydrogen fuel into a combustion chamber of the internal combustion engine. By way of example, the fluid management system may comprise a housing, wherein a pipe is connected to a connection point portion by means of the pipe connection arrangement. In addition, the fuel injector assembly is accommodated in the housing, wherein a fluid port of an injector is arranged in fluid communication with the connection point portion.

(37) In one example, the fluid management system may comprise a housing for accommodating an injector of the fuel injector assembly, and the pipe connection arrangement may be configured to connect the pipe to the connection point portion, whereby the fluid port of the injector may be arranged in fluid communication with the connection point portion.

(38) According to a third aspect of the present disclosure, there is provided a vehicle comprising a fluid management system according to any one of the example embodiments described in relation to the second aspect and/or a pipe connection arrangement according to any one of the example embodiments described in relation to the first aspect. The vehicle may be a truck, such as a heavy-duty truck. However, the disclosure is not limited to trucks and heavy-duty trucks. Rather, the vehicle may be any type of vehicle, such as a truck, car, working machine intended for carrying out

an operation etc.

(39) Effects and features of the third aspect are largely analogous to those described above in relation to the first aspect and/or the second aspect.

(40) The fluid management system may be an integral part of an internal combustion engine. Alternatively, the fluid management system may be a separate part connected to the internal combustion engine.

(41) Further features of, and advantages with, the present disclosure will become apparent when studying the appended claims and the following description. The skilled person will realize that different features of the present disclosure may be combined to create embodiments other than those described in the following, without departing from the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above, as well as additional objects, features and advantages of the present disclosure, will be better understood through the following illustrative and non-limiting detailed description of exemplary embodiments of the present disclosure, wherein:

(2) FIG. 1 is a lateral side view illustrating an example of a vehicle in the form of a truck; the vehicle comprising an internal combustion engine system having a fluid management system and a pipe connection arrangement according to one example embodiment of the present disclosure;

(3) FIG. 2 schematically illustrates parts of the engine in FIG. 1 comprising an example embodiment of a fluid management system and a pipe connection arrangement according to the present disclosure;

(4) FIGS. 3A to 3C schematically illustrate an example embodiment of the fluid management system and the pipe connection arrangement according to the present disclosure;

(5) FIG. 3D schematically illustrate another example embodiment of the fluid management system and the pipe connection arrangement according to the present disclosure;

(6) FIG. 4A is an exploded perspective view of the fluid management system and the pipe connection arrangement in FIGS. 3A to 3C;

(7) FIG. 4B schematically illustrates one component of the pipe connection arrangement in FIGS. 3A to 3C; and

(8) FIGS. 5A and 5B schematically illustrate a sequence of connecting a pipe to a fluid management system by means of the pipe connection arrangement in FIGS. 3A to 3C.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS OF THE DISCLOSURE

(9) The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the disclosure are shown. The disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided for thoroughness and completeness. Like reference character refer to like elements throughout the description.

(10) With particular reference to FIG. 1, there is provided a vehicle 1 in the form of a truck. The vehicle comprises an internal combustion engine system 4 having an internal combustion engine 6, such as a diesel engine. Further, the engine or engine system comprises a fluid management system 8 according to any one of the example embodiments of the present disclosure. The fluid management system 8 comprises a pipe connection arrangement 10 according to any one of the example embodiments of the present disclosure. In the following FIGS. 2, 3A to 3D, 4A to 4B and 5A to 5B, some examples of the pipe connection arrangement 10 will be described in greater detail.

(11) FIG. 2 schematically illustrates an example embodiment of parts of the fluid management system 8 installed to the internal combustion engine 6 in FIG. 1. The internal combustion engine 6

comprises one or more cylinders **6a** and an intake manifold **6b** in fluid communication with the cylinder. The internal combustion engine **6** further comprises an intake valve **6c** arranged in the intake manifold for regulating the flow of compressed air to a combustion chamber **6d** of the internal combustion engine **6**.

(12) As illustrated in FIG. 2, the fluid management system **8** here comprises an injector **8a**. Hence, the fluid management system **8** is here provided in the form of an injector assembly comprising the injector **8a**. The injector **8a** is configured to inject a fuel for combustion in the combustion chamber **6a**. Thus, the injector **8a** is a fuel injector. In FIG. 2, the fluid management system **8** is a so called port-injector fluid management system. In such system, the injector **8a** is arranged to inject fuel into the intake manifold **6b** upstream the intake valve **6b**. Other arrangements of the injector **8a** are also conceivable. For instance, the injector may be arranged in a cylinder head of the cylinder, through which fuel is injected into the cylinder **6a** as a fuel spray.

(13) The fuel is here petrol fuel. In other examples, the fuel is hydrogen fuel. In yet other examples, the fuel may be diesel fuel. For diesel ICE systems, the fuel is preferably injected with a pressure in the range 600 to 3000 bar. Generally, for an engine system using EGR, about 1000 to 2500 bar may be preferred, without EGR about 800 to 1400 bar. For hydrogen ICE systems, the hydrogen gas fuel may be injected with a low injection pressure of between 15 to 60 bar into the combustion chamber **6d** and towards the piston. However, for other gaseous ICE systems, the fuel injector may be controllable to inject gaseous fuel into the combustion chamber with an injection pressure of up to about 500 bar. As such, there is a need for connecting the fuel pipe **12** to the injector **8a** in a secure manner.

(14) By way of example, the fuel injector **8a** is connected and in fluid communication with a fuel tank (not illustrated). The number of fuel injectors may be equal to the numbers of cylinders of the ICE. The fuel injectors are each arranged in fluid communication with the fuel tank. Each one of the fuel injectors is connected to the fuel tank by a corresponding pipe **12**, as illustrated in FIG. 2. In addition, as depicted in FIG. 2, the pipe **12** is connected to the fluid management system **8**, depicted as an injector assembly, by means of a pipe connection arrangement **10**. In FIG. 2, the pipe connection arrangement **10** is configured to releasably connect an end portion **9** of the pipe **12** to a fluid port of the fluid management system **8**, i.e. the injector assembly. The fluid port may either be an integral part of the injector **8a** or a fluid port of the injector assembly that is in fluid communication with the injector **8a**.

(15) The parts of the fluid management system **8** as well as the pipe connection arrangement will now be described in more detail in relation to the FIGS. 3A to 3C, 4A to 4B and 5A to 5B. FIG. 3D is another example of a pipe connection arrangement **10** and a fluid management system **8** for use in the vehicle in FIG. 1.

(16) The fluid management system **8** and the pipe connection arrangement **10** will be described in an orientation to the ICE **6**, as illustrated in e.g. FIGS. 3A to 3C. However, the orientation of the fluid management system **8** and the pipe connection arrangement **10** as depicted in e.g. FIG. 3A is merely used as an example for ease of understanding the attachment of the pipe **12** to the fluid management system **8**, and other attachment arrangements and orientations of the fluid management system **8** and the pipe connection arrangement **10** may be conceivable. As indicated in e.g. FIG. 3A, the fluid management system **8** has an extension in a longitudinal direction X, an extension in a transverse direction Y and an extension in a vertical direction Z. Analogously, the components of the pipe connection arrangement **10** have an extension in a longitudinal direction X, an extension in a transverse direction Y and an extension in a vertical direction Z. Some of the components may also have a circular cross-section and a cylindrical shape. These types of components may also be defined as having an extension in an axial direction A, an extension in a radial direction R and an extension in a circumferential direction C. Generally, as illustrated in e.g. FIG. 3A, when the pipe **12** is connected to the fluid management system **8** by the pipe connection arrangement **10**, the direction Z corresponds to the axial direction A, as will be further described

herein in relation to the components making up the pipe connection arrangement **10**.

(17) Turning now to FIGS. **3A** to **3C**, there is depicted one example embodiment of the pipe connection arrangement **10** for the fluid management system **8** in FIG. **2**. FIG. **3A** is a perspective view of the pipe connection arrangement **10** and the fluid management system **8**, in which the pipe **12** is connected to the fluid management system **8** by means of the pipe connection arrangement **10**. FIG. **3B** is a cross-sectional view of the pipe connection arrangement **10** and the fluid management system **8**, as seen in a cross-sectional plane extending in the longitudinal direction X and the vertical direction Z. In FIG. **3B**, the pipe **12** is connected to the fluid management system **8** by means of the pipe connection arrangement **10**. FIG. **3C** is another cross-sectional view of the pipe connection arrangement **10** and the fluid management system **8**, as seen in a cross-sectional plane extending in the longitudinal direction X and the transverse direction Y, and in which the pipe **12** is connected to the fluid management system **8** by means of the pipe connection arrangement **10**.

(18) As illustrated in FIGS. **3A** to **3C**, the fluid management system **8** comprises a housing **8b** for the injector **8a**. In addition, the housing **8b** comprises an outer recess **8c** for the pipe **12**. The outer recess **8c** defines a restricted space for any tools needed for connecting and disconnecting the pipe **12** to the housing **8b**.

(19) The fluid management system **8** comprises the pipe connection arrangement for connecting the pipe **12** to the fluid port **2**, as illustrated e.g. in FIG. **3B**. One example of the pipe connection arrangement will now be described with reference to FIGS. **3A** to **3C** in conjunction with FIG. **4A**. FIG. **4A** is an exploded perspective view of the fluid management system **8** and the pipe connection arrangement **10** in FIGS. **3A** to **3C**.

(20) As illustrated in e.g. FIG. **3C** in combination with FIG. **4A**, the pipe connection arrangement **10** comprises a pipe nut **11**. The pipe nut **11** has a first portion **13**, which is here provided as a head portion. The first portion **13** defines an outer side and an inner side. The first portion **13** is thus generally provided in the form of a ring-shaped portion, resembling a conventional screw nut. As illustrated in e.g. FIG. **3C**, the first portion **13** comprises a toothed outer profile **14**. The toothed outer profile **14** may be considered as a first outer profile **14** of the pipe nut **11**.

(21) The pipe nut **11** defines an inner hole for receiving the end portion **9** of the pipe **12**. In this example, the pipe nut **11** is arranged about the pipe **12**. The pipe nut **11** is arranged around the pipe **12** such that the end portion **9** of the pipe **12** is located outside the pipe nut **11**. That is, the pipe nut **11** is arranged circumferentially around the pipe **12**, whilst the end portion **9** of the pipe **12** projects away from the pipe nut **11** in the axial direction A of the pipe **12**, here also corresponding to the vertical direction Z.

(22) The pipe nut **11** also comprises a second portion **15**. The second portion **15** is configured to engage with the connection point portion **4** of the fluid management system **8**. The connection point portion **4** is here in fluid communication with the fluid port **2** of the injector **8a**. As such, the second portion **15** is configured to indirectly connect with the fluid port **2** of the injector **8a** of the fluid management system **8** so as to provide a fluid connection therebetween. In this example, the second portion **15** comprises an outer threaded profile **16**, as depicted in FIG. **4A**. The outer threaded profile **16** may be considered as a second outer profile of the pipe nut **11**. The second portion **15** thus comprises the second outer profile of the pipe nut **11**. In other examples, although not illustrated, the second portion may be an inner threaded of the pipe nut. The second portion may in addition, or alternatively, be a bayonet connection portion. In such example, the connection point portion **4** generally comprises a corresponding bayonet connection portion.

(23) By way of example, the outer threaded profile **16** is a conventional thread, also commonly denoted as a screw thread. The screw thread is defined as a ridge of uniform section in the form of a helix on the surface of the pipe nut **11**. The connection point portion **4** generally comprises a corresponding threaded portion (not illustrated) configured to engage with the outer threaded profile **16**.

(24) To this end, by the first portion **13** and the second portion **15**, the pipe nut **11** is configured to releasably connect the end portion **9** of the pipe **12** to the connection point portion **4** of the fluid management system **8**, and thus to connect the pipe **12** in fluid communication with the fluid port **2** of the injector **8a**. By this connection configuration, the arrangement of the pipe nut **11** is configured to clamp and retain, in a fluid-sealing manner, the pipe **12** to the connection point portion **4** of the fluid management system **8**. In other examples, the pipe nut **11** is configured to releasably connect the end portion **9** of the pipe **12** directly to a connection point portion **4** in the form of the fluid port **2** of the injector **8a**. The connection point portion is here an integral part of the injector. However, the connection point portion may in other examples be an additional part and/or a separate part from the injector. Hence, the connection point portion **4** may be an interconnecting portion arranged to interconnect the pipe **12** with the injector **8a**.

(25) Moreover, as illustrated in e.g. FIG. 4A, the pipe connection arrangement **10** here comprises an elongated drive member **20**. The elongated drive member **20** is a tool component used at the times of assembly and disassembly of the pipe **12** to the fluid management system **8**, but generally not during transportation of fluid through the pipe **12** to the fluid management system **8**. The elongated drive member **20** extends in the axial direction A and in the radial direction R. The elongated drive member **20** further has a centre axis Ac, extending in the axial direction A. The axial direction is parallel to the vertical direction Z. The centre axis Ac of the elongated drive member **20** here corresponds to the axis of rotation of the elongated drive member **20**. FIG. 4B illustrates further parts of the elongated drive member **20** in FIG. 4A. As illustrated in FIG. 4B, the elongated drive member **20** has an essentially cylindrical shape with a substantial length in the axial direction A.

(26) The elongated drive member **20** comprises a first end portion **21**. The first end portion **21** comprises a toothed outer profile **22**. The toothed outer profile **22** is configured to cooperate with the toothed outer profile **14** of the pipe nut **11**. In addition, the toothed outer profile **22** and the toothed outer profile **14** are complementary in shape so as to provide for a functional interaction therebetween upon rotation of the elongated drive member **20**, as illustrated in the FIGS. 5A and 5B.

(27) In this example, the toothed outer profile **22** of the elongated drive member **20** has a six-point star-shaped pattern. The six-point star-shaped pattern is here provided in the form of a TORX-profile. Accordingly, the toothed outer profile **22** is configured to cooperate with the toothed outer profile **14** of the pipe nut **11**.

(28) As such, the elongated drive member **20** is arranged and configured to engage with the toothed outer profile **14** of the pipe nut **11**, as also depicted in e.g. FIG. 3C.

(29) As illustrated in FIG. 4B, the elongated drive member **20** comprises a second end portion **24**. The second end portion **24** is arranged at an opposite axial end of the elongated drive member **20**. That is, the first and second end portions **21**, **24** are opposite axial end portions of the elongated drive member **20**. The second end portion **24** is a head portion of the elongated drive member **20**. The second end portion **24** is adapted to cooperate with a tool, such as a wrench, a screwdriver or the like. By way of example, the second end portion **24** is a hexagon head portion. Other designs of the second end portion may also be conceivable as long as the elongated drive member **20** can be manipulated by a tool. The second end portion **24** and the first end portion **22** may also be distanced from each other by means of an intermediate cylindrical part extending in the axial direction and between the first and second end portions **22**, **24**.

(30) Further, the pipe connection arrangement **10** comprises a supplementary guide member **30**, as illustrated in FIGS. 3A to 3C, and also in FIG. 4A. The supplementary guide member **30** has a corresponding centre axis Ac1. The supplementary guide member **30** can be provided in several different shapes and configurations. In this example, the supplementary guide member **30** is a guide pin. Moreover, the supplementary guide member **30** is adapted for arrangement in the fluid management system **8** at a position adjacent the pipe connection point portion **4**. Thus, the

supplementary guide member **30** is associated with the fluid port **2**. The supplementary guide member **30** comprises an outer end part **31**. As illustrated in e.g. FIG. **4A**, the outer end part **31** defines a cylindrical shaped portion. Accordingly, the supplementary guide member **30** is here at least partly of a cylindrical shape. Thus, the supplementary guide member **30** has a circular cross section and a substantial extension in the axial direction **A**. However, the supplementary guide member may also be provided in other designs, e.g. with a rectangular cross section and a substantial length in the axial direction **A**.

(31) In FIGS. **3A** to **3C**, the supplementary guide member **30** is a separate part attached to the fluid management system **8**. As such, the supplementary guide member **30** is adapted to be fixedly arranged adjacent the connection point portion **4** of the fluid management system **8**. In other examples, the supplementary guide member **30** is an integral part of the fluid management system **8**.

(32) By way of example, the supplementary guide member **30** is a cylindrical projecting segment **32**, as illustrated in FIGS. **3A** to **3C**. In other examples, the supplementary guide member **30** may be a cylindrical recessed segment **33**, as illustrated in FIG. **3D**. In FIG. **3D**, the supplementary guide member **30** is provided in the form of a cylindrical recessed segment **33** with an inner diameter **34**. The cylindrical projecting segment and/or cylindrical recessed segment may be machined in the fluid management system **8**.

(33) The elongated drive member **20** and the supplementary guide member **30** are arranged and configured to engage with each other. By way of example, the first end portion **21** is configured to concentrically engage with the supplementary guide member **30**. By way of example, as depicted in FIG. **4B**, the first end portion **21** comprises a recess **23**. The recess **23** is arranged concentrically about the centre axis **Ac** of the elongated driver member **20**. The recess **23** is further adapted to at least partly encompass the outer end part **31** of the supplementary guide member **30**.

(34) As depicted in FIG. **4B**, the recess **23** is a cylindrical recess. The cylindrical recess **23** is arranged concentric with the first end portion **21**. In addition, the cylindrical recess **23** extends from an end face of the first end portion **21**.

(35) Furthermore, a diameter of the cylindrical recess **23** is here essentially equal to a diameter of a central recess of a so called Security TORX® screwdriver bit of the appropriate standard size. Security TORX®, or tamper proof TORX screwdriver bits have the exact same exterior shape as regular TORX® screwdriver bits, but have a small hole drilled in the end to accept the shape and geometry of the supplementary guide member **30**. However, other shapes and diameters may be selected for other types of elongated drive members.

(36) It is to be noted that the supplementary guide member **30** may be designed with a tolerance relative the cylindrical recess **23**. Hence, in FIGS. **3A** to **3C**, the cylindrical recess **23** and the supplementary guide member **30** form a locational clearance fit upon concentric engagement of the supplementary guide member **30** and the elongated drive member **20**.

(37) As mentioned above, FIG. **3D** schematically illustrates another example of the design of the supplementary guide member **30**. In this example, the supplementary guide member is a recessed segment **33** formed by a cylindrical part having an internal diameter **34**. The cylindrical recessed segment **33** is here a part of the fluid management system **8**. Hence, the fluid management system **8** comprises the cylindrical recessed segment **33** having the internal diameter **34** (rather than the projecting segment as illustrated in FIG. **3C**). As illustrated in FIG. **3D**, the size of the internal diameter **34** of the cylindrical recessed segment **33** is set such that upon the concentric engagement of the elongated drive member **20** with the supplementary guide member **30**, the internal diameter **34** of the cylindrical recessed segment **33** forms a locational clearance fit with a maximum diameter of the toothed outer profile **22**. Unless explicitly mentioned herein, the example in FIG. **3D** can include any other feature, example and/or aspect as mentioned herein with respect to FIGS. **1**, **2**, **3A** to **3C**, **4A** to **4B** and **5A** to **5B**.

(38) As illustrated in FIG. **3B** in conjunction with FIG. **3C**, the supplementary guide member **30** is

positioned at a distance from the connection point portion **4**. By way of example, the supplementary guide member **30** is positioned at a transverse distance from the connection point portion **4**. The supplementary guide member **30** may in addition be positioned at a longitudinal distance from the connection point portion **4**. Further, the supplementary guide member **30** may be positioned at a distance from the connection point portion **4** as seen in the transverse and longitudinal directions.

(39) Turning again to the pipe nut **11** and FIG. 3C, the pipe nut **11** has a centre axis **Ac2**. The centre axis **Ac2** of the pipe nut **11** here corresponds to the axis of rotation of the pipe nut **11**. Thus, the pipe nut **11** has an axis of rotation, **Ac2**. As illustrated in FIG. 3C, when the pipe nut **11** is in threaded engagement with the thread of the connection point portion **4**, the axis of rotation **Ac2** of the pipe nut **11** is essentially parallel to the centre axis **Ac1** of the supplementary guide member **30**, i.e. to the cylindrical part of the supplementary guide member **30**.

(40) As further illustrated in FIG. 3C, when the elongated drive member **20** is in engagement with the supplementary guide member **30**, and the second portion **15** of the pipe nut **11** is in engagement with the connection point portion **4**, the toothed outer profile **22** and the toothed outer profile **14** of the pipe nut **11** are permitted to mesh together. In this manner, a rotation of the elongated drive member **20** causes a corresponding rotation of the pipe nut **11** for clamping the end portion **9** to the fluid port **2** or unclamping the end portion **9** from the fluid port **2**.

(41) By the above arrangement, it thus becomes possible to effect a connection of the pipe **12** to the connection point portion **4** by manipulating the elongated drive member **20** and the pipe nut **11**.

(42) More specifically, when the toothed outer profile **22** of the elongated drive member **20** is engaged with the toothed outer profile **14** of the pipe nut **11**, a rotation of the elongated drive member **20** about its centre axis **Ac** causes the toothed outer profile **22** of the elongated drive member **20** to effect a corresponding rotation of the pipe nut **11** about its centre axis **Ac2**, thereby effecting a tightening or loosening of the pipe **12** relative to the pipe connection point portion **4**. In this example, as illustrated in FIG. 4A, the pipe connection point portion **4** is arranged on the housing **8b** of the injector assembly (fluid management system **8**), whilst the fluid port **2** is arranged on the injector **8a**. The fluid port **2** is in fluid communication with the connection point portion **4**. In other examples, the fluid port **2** corresponds to the pipe connection point portion **4**.

(43) As mentioned above, and illustrated in e.g. FIG. 3C, the supplementary guide member **30** is arranged adjacent the pipe nut **11**. In addition, the corresponding centre axis **Ac** of the elongated drive member **20** and the centre axis of the **Ac2** of the pipe nut **11** are here arranged parallel to each other. In other words, the corresponding centre axis **Ac** of the elongated drive member **20** and the centre axis of the **Ac2** of the pipe nut **11** are on the same plane. Thus, the corresponding centre axis **Ac** of the elongated drive member **20** and the centre axis of the **Ac2** of the pipe nut **11** are arranged adjacent each other as seen in a plane extending perpendicular to the corresponding centre axis **Ac1** of the supplementary guide member **30** and the axis of rotation (centre axis) **Ac2** of the pipe nut **11**. Moreover, when the first end portion **21** of the elongated drive member **20** is engaged with the supplementary guide member **30**, the elongated drive member **20** is arranged adjacent the pipe nut **11** as seen in a plane extending perpendicular to the corresponding centre axis **Ac** of the elongated drive member **20** and the axis of rotation **Ac2** of the pipe nut **11**.

(44) Typically, although strictly not required, the pipe connection arrangement **10** comprises an additional supplementary guide member **30'** and an additional elongated drive member **20'**, as illustrated in FIGS. 3A to 3C and FIGS. 4A to 4B. In other words, the pipe connection arrangement **10** comprises the additional supplementary guide member **30'**. The additional supplementary guide member **30'** is here essentially identical in design to the supplementary guide member **30**. Thus, the supplementary guide member **30** is a first supplementary guide member **30** and the additional supplementary guide member **30'** is a second supplementary guide member.

(45) As illustrated in FIG. 3C, the additional supplementary guide member **30'** is arranged asymmetrical from the position of the first supplementary guide member **30** relative to the axis of

rotation Ac2 of the pipe nut **11**.

(46) In addition, as illustrated in FIG. 3C, the additional supplementary guide member **30'** is positioned at an essentially same transverse, or lateral, distance from the connection point portion **4** as the first supplementary guide member **30**, but at different angular location around a circumference of the connection point portion **4** relative to the angular location of the first supplementary guide member **30**.

(47) By way of example, as illustrated in FIG. 3C, the additional supplementary guide member **30'** is positioned at an angle α being greater than 120 degrees relative to the supplementary guide member **30**. As illustrated in FIG. 3C, the angle α is the sum of the α_1 and α_2 . Such arrangement may at least partly overcome tilting of the pipe nut in its thread that increases friction in the threads and reduces useful torque. In practice, one supplementary guide member **30** may generally be used for initial tightening and then the additional supplementary guide member **30'** for additional tightening. Positioning the additional supplementary guide member **30'** more towards the opposite side of the pipe nut **11** may help to counter the tilt introduced when using the supplementary guide member **30**.

(48) In addition, or alternatively, the additional supplementary guide member **30'** is positioned at an angular location that is not a multiple of 360 degrees divided by the number of teeth of the toothed outer profile **14**. Such arrangement allows for using a less complex shaped pattern of the toothed profile that generally require more than one relative position of the elongated drive member **20** and the pipe nut **11** for efficient enough force transfer. In practice, the elongated drive member **20** may occasionally slip relative to the pipe nut **11** once a particular angular position of the pipe nut **11** is reached. In such cases, the elongated drive member **20** is moved to the other supplementary guide member **30** (to continue tightening the pipe nut) where the supplementary guide member **30** may not slip.

(49) In some designs, as illustrated in FIGS. 3A to 3C and FIGS. 4A to 4B, the pipe connection arrangement **10** comprises an additional elongated drive member **20'** having a corresponding first end portion **21'** comprising a corresponding outer profile **22'**. The outer profile **22'** is configured to mesh with the toothed outer profile **14** upon engagement with and subsequent rotation of the additional elongated drive member **20'**. The corresponding first end portion **21'** is configured to at least partly encompass a corresponding outer end part **31'** of the additional supplementary guide member **30'**. The corresponding first end portion **21'** may further have a corresponding recess, as the recess **23**, which is configured to at least partly encompass an outer end part **31'** of the additional supplementary guide member **30'**.

(50) FIGS. 5A and 5B schematically illustrate a sequence of connecting the pipe **12** to the fluid management system **8** by means of the pipe connection arrangement **10** in FIGS. 3A to 3C and FIGS. 4A to 4B.

(51) In FIG. 5A, the pipe connection arrangement **10** and the pipe **12** are illustrated in a state in which the pipe **12** is disconnected, and/or detached, from the connection point portion **4** of the fluid management system **8**. In FIG. 5B, the pipe connection arrangement **10** and the pipe **12** are illustrated in a state in which the pipe **12** is connected and clamped to the connection point portion **4** of the fluid management system **8**, thereby forming a liquid-tight configuration for the fluid. By the pipe connection arrangement **10**, the pipe **12** is securely connected to the fluid management system **8** so that fluid can be transported through the pipe **12** to the injector **8a**.

(52) More specifically, in a first assembly state of the pipe **12** to the fluid management system **8**, as shown in FIG. 5A and indicated by reference numeral **110**, the pipe **12** with the pipe nut **11** is inserted into the connection point portion **4** of the fluid management system **8**. The pipe **12** with the pipe nut **11** is inserted into the connection point portion **4** of the fluid management system **8** along the vertical direction Z, generally corresponding to the axial direction A of the pipe nut **11** and the pipe **12**.

(53) The pipe nut **11** with the pipe **12** is inserted into a hole of the fluid management system **8**,

defining the connection point portion **4**, until the end portion **9** is at least partly positioned within the fluid management system **8**.

(54) Subsequently, in a second assembly state, as indicated by reference numeral **120**, each one of the elongated drive members **20** and **20'**, are moved in the axial direction A, generally corresponding to the vertical direction Z, towards the corresponding supplementary guide members **30**, **30'**. As mentioned above, when the recess **23** of the first end portion **21** at least partly encircles or accommodate the outer end part **31** of the supplementary guide member **30**, the elongated drive member **20** is arranged in engagement with the supplementary guide member **30**. Analogously, when the corresponding recess of the corresponding first end portion **21'** at least partly encircles or accommodates the corresponding outer end part **31'** of the additional (second) supplementary guide member **30'**, the additional elongated drive member **20'** is arranged in engagement with the additional (second) supplementary guide member **30'**. In this state, the respective elongated drive members **20**, **20'** are arranged to be rotatable about corresponding outer end parts **31**, **31'**.

(55) Moreover, due to the relative arrangement of the supplementary guide member **30**, the elongated drive member **20** and the pipe nut **11** (which is in engagement with the connection point portion **4**), the toothed outer profile **22** and the toothed outer profile **14** of the pipe nut **11** are arranged to mesh with each other. Analogously, due to the relative arrangement of the additional supplementary guide member **30'**, the additional elongated drive member **20'** and the pipe nut **11** (which is in engagement with the connection point portion **4**), the toothed outer profile **22'** and the toothed outer profile **14** of the pipe nut **11** are arranged to mesh with each other from another side of the pipe nut **11**.

(56) Accordingly, in the second assembly state **120**, the pipe nut **11** is in the threaded engagement with the thread of the connection point portion **4**, wherein the toothed outer profile **14** of the pipe nut **11** and the toothed outer profile **22** of the elongated drive member **20** mesh together such that a rotation of the elongated drive member **20** results in a corresponding rotation of the pipe nut **11** for clamping the end portion **9** of the pipe **12** to, or un-clamping it from, the fluid port **2**. The rotations of the respective elongated drive members **20**, **20'** are schematically illustrated in FIG. 5B, and corresponds to a third, and final, assembly state **130**. As such, during a subsequent rotation of the elongated drive member **20** in its circumferential direction, and when the elongated drive member **20** is engaged with the supplementary guide member **30**, the toothed outer profile **22** of the elongated drive member **20** engages and meshes with the outer profile **14** of the pipe nut **11** so as to effect a tightening of the pipe **12** relative to the pipe connection point portion **4** upon a rotation in a first circumferential direction, e.g. in a clockwise direction, or a loosening of the pipe **12** relative to the pipe connection point portion **4** upon a rotation in a second and opposite circumferential direction, e.g. in a counter-clockwise direction.

(57) As may be gleaned from e.g. FIG. 5B, but also illustrated in the other Figures. e.g. FIGS. 3A to 3C, the extensions of the components of the pipe connection arrangement **10** in the axial direction A generally coincide with the vertical extension Z of the fluid management system **8** when the pipe **12** is inserted and connected to the connection point portion **4** of the fluid management system **8**. Thus, as depicted in FIG. 5B, the components of the pipe connection arrangement and the end part **9** of the pipe **12** are essentially oriented in parallel with the vertical direction Z of the fluid management system **8**.

(58) It should also be readily appreciated that the elongated drive members **20**, **20'** are only parts of the pipe connection arrangement **10** during installation and removal of the pipe **12** to the connection point portion **4** of the fluid management system.

(59) Moreover, it should be readily appreciated that when the pipe end **9** is loosened, it may remain in almost the same position as if it were tightened, i.e. about the pipe **12** as illustrated in e.g. FIG. 4A.

(60) As mentioned above, it should be readily appreciated that the pipe connection arrangement **10** can be provided using merely one elongated drive member **20** and one supplementary guide

member **30**. Equally, the pipe connection arrangement **10** can be provided by applying one elongated drive member **20** over two supplementary guide members **30**, **30'** in an alternating manner. Upon finalizing the connection and tightening up of the pipe nut **11**, the elongated drive member tool is removed until pipe disconnection or re-tightening is needed at a later time. Hence, the example with two elongated drive members **20**, **20'** and two supplementary guide members **30**, **30'** are merely an example for illustrating the functionality of the pipe connection arrangement **10**. (61) To sum up, there is provided a pipe connection arrangement **10** for the fluid management system **8**. The pipe connection arrangement **10** comprises the pipe nut **11** for releasably connecting the end portion **9** of the pipe **12** to the fluid port **2** of the fluid management system **8**. The pipe nut comprises the first portion **13** having the toothed outer profile **14** and the second portion **15** configured to engage with the connection point portion **4** of the fluid management system **8**. The pipe connection arrangement **10** also comprises the elongated drive member **20** having the first end portion **21** with the toothed outer profile **22**. The toothed outer profile **22** is configured to cooperate with the toothed outer profile **14** of the pipe nut **11**. The first end portion **21** is further configured to concentrically engage with the supplementary guide member **30**, which is adapted to be fixedly arranged adjacent the connection point port **4** of the fluid management system **8**. Also, when the elongated drive member **20** is in engagement with the supplementary guide member **30**, and the second portion **15** of the pipe nut **11** is in engagement with the connection point portion **4**, the toothed outer profile **22** and the toothed outer profile **14** of the pipe nut **11** are permitted to mesh together such that a rotation of the elongated drive member **20** causes a corresponding rotation of the pipe nut **11** so as to either clamp the end portion **9** to the fluid port **2** or unclamping the end portion from the fluid port **2**.

(62) In addition, the present disclosure relates to the fluid management system **8** comprising the pipe connection arrangement **10** according to any one of the examples described and illustrated herein.

(63) Further, the present disclosure relates to the vehicle **1** comprising any one of the pipe connection arrangements **10** and/or the fluid management system **8** according to any one of examples described and illustrated herein.

(64) The pipe connection arrangement **10** is here made of steel, such as stainless steel, or any other suitable metallic material. However, the pipe connection arrangement **10** may be made of a lightweight material, such as a plastic material, lightweight steel, aluminium or similar material.

(65) The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(66) It will be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element without departing from the scope of the present disclosure.

(67) Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “transversal” or “vertical” may be used herein to describe a relationship of one element to another element as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the system in addition to the orientation depicted in the Figures. Hence, it should be noted that the terms top and bottom, upper and lower, as well as any other similar terms are used in reference to the position of the fluid management

system 8 as depicted in the drawings, e.g. FIGS. 3A to 5B, and the system may be positioned and used in other orientations.

(68) It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element, or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

(69) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(70) Also, it should be readily appreciated that although some components are mentioned and/or illustrated as having certain dimensions and shapes, the dimensions and shapes of these components may likewise be provided in other shapes and dimensions, such as components having cross sections resembling circular-, oval-, rectangular- and triangular-shaped cross sections, or any other conceivable cross-section, as long as nothing else is mentioned in relation to these components and the components can provide their functions.

(71) It is to be understood that the present disclosure is not limited to the aspects described above and illustrated in the drawings; rather, the skilled person will recognize that many changes and modifications may be made within the scope of the present disclosure and appended claims. In the drawings and specification, there have been disclosed aspects for purposes of illustration only and not for purposes of limitation, the scope of the inventive concepts being set forth in the following claims.

Claims

1. A pipe connection arrangement for a fluid management system, said pipe connection arrangement comprising: a pipe nut for releasably connecting an end portion of a pipe to a fluid port of the fluid management system, said pipe nut comprising a first portion having a toothed outer profile and a second portion configured to engage with a connection point portion of said fluid management system, an elongated drive member having a first end portion with a toothed outer profile being configured to cooperate with the toothed outer profile of the pipe nut, said first end portion further being configured to concentrically engage with a supplementary guide member adapted to be fixedly arranged adjacent the connection point portion of said fluid management system, whereby, when the elongated drive member is in engagement with the supplementary guide member, and the second portion of the pipe nut is in engagement with the connection point portion, the toothed outer profile of the elongated drive member and the toothed outer profile of the pipe nut are permitted to mesh together such that a rotation of said elongated drive member causes a corresponding rotation of the pipe nut for clamping the end portion to the fluid port or unclamping the end portion from the fluid port, wherein said first end portion comprises a recess arranged concentrically about a centre axis of the elongated drive member and further adapted to at least partly encompass an outer end part of the supplementary guide member, wherein, when said first end portion of the elongated drive member is engaged with the supplementary guide member, said elongated drive member being arranged adjacent the pipe nut, and corresponding centre axis of said elongated drive member and said pipe nut, respectively, being arranged parallel to each other.

2. Pipe connection arrangement according to claim 1, wherein said pipe connection arrangement further comprises the supplementary guide member, said supplementary guide member being positioned at a distance from the connection point portion such that an axis of rotation of the pipe nut, when in engagement with the connection point portion, is essentially parallel to a centre axis of

the supplementary guide member.

3. Pipe connection arrangement according to claim 1, further comprising an additional supplementary guide member, said additional supplementary guide member being positioned at an essentially same transverse distance from the connection point portion as the supplementary guide member, but at different angular location around a circumference of the connection point portion relative to the angular location of the supplementary guide member.

4. Pipe connection arrangement according to claim 3, wherein the additional supplementary guide member is positioned at an angular location that is not a multiple of 360 degrees divided by the number of teeth of the toothed outer profile.

5. Pipe connection arrangement according to claim 3, wherein the additional supplementary guide member is positioned at an angle being greater than 120 degrees relative to the supplementary guide member.

6. Pipe connection arrangement according to claim 3, wherein said pipe connection arrangement comprises an additional elongated drive member having a corresponding first end portion comprising a corresponding outer profile configured to mesh with the toothed outer profile upon engagement with and subsequent rotation of the additional elongated drive member, said corresponding first end portion being configured to at least partly encompass a corresponding outer end part of the additional supplementary guide member.

7. Pipe connection arrangement according to claim 1, wherein the second portion is either an outer threaded profile or an inner threaded profile of the pipe nut.

8. Pipe connection arrangement according to claim 1, wherein the toothed outer profile of the elongated drive member defines a six-point star-shaped pattern.

9. A fluid management system comprising a pipe connection arrangement according to claim 1.

10. Fluid management system according to claim 9, wherein the supplementary guide member is a separate part attached to the fluid management system.

11. Fluid management system according to claim 9, wherein the supplementary guide member is comprised with the fluid management system, and provided in the form of a cylindrical projecting segment or a cylindrical recessed segment with an inner diameter.

12. Fluid management system according to claim 9, said fluid management system comprising a fuel injector assembly for an internal combustion engine.

13. Fluid management system according to claim 12, wherein the fluid management system comprises a housing for accommodating an injector of the fuel injector assembly, the pipe connection arrangement being configured to connect the pipe to the connection point portion, whereby a fluid port of the injector is arranged in fluid communication with the connection point portion.

14. A vehicle comprising any one of a pipe connection arrangement according to claim 1.

15. A fluid management system, comprising: a pipe connection arrangement comprising: a pipe nut for releasably connecting an end portion of a pipe to a fluid port of the fluid management system, said pipe nut comprising a first portion having a toothed outer profile and a second portion configured to engage with a connection point portion of said fluid management system, an elongated drive member having a first end portion with a toothed outer profile being configured to cooperate with the toothed outer profile of the pipe nut, said first end portion further being configured to concentrically engage with a supplementary guide member adapted to be fixedly arranged adjacent the connection point portion of said fluid management system, whereby, when the elongated drive member is in engagement with the supplementary guide member, and the second portion of the pipe nut is in engagement with the connection point portion, the toothed outer profile of the elongated drive member and the toothed outer profile of the pipe nut are permitted to mesh together such that a rotation of said elongated drive member causes a corresponding rotation of the pipe nut for clamping the end portion to the fluid port or unclamping the end portion from the fluid port, wherein said first end portion comprises a recess arranged concentrically about a

center axis of the elongated drive member and further adapted to at least partly encompass an outer end part of the supplementary guide member; wherein the fluid management system comprises a fuel injector assembly for an internal combustion engine.

16. The fluid management system according to claim 15, wherein the fluid management system comprises a housing for accommodating an injector of the fuel injector assembly, the pipe connection arrangement being configured to connect the pipe to the connection point portion, whereby the fluid port of the injector is arranged in fluid communication with the connection point portion.
